

Assignment 1 — AI Ethics and Bias Analysis

Course: ITAI 4376 — AI Applications

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Problem Statement

This assignment examined the ethical challenges associated with deploying artificial intelligence systems, with particular attention to algorithmic bias, fairness, and accountability. The central question was: how do biases enter AI systems, what harm can they cause, and what frameworks exist to address them?

Approach and Methods

The analysis began by identifying the primary sources of bias in machine learning pipelines, including biased training data, flawed feature selection, and lack of diversity in development teams. I reviewed documented cases of AI bias in hiring algorithms, facial recognition systems, and clinical decision support tools to illustrate how these issues manifest in practice.

I then examined existing ethical frameworks and regulatory guidelines — including the EU AI Act and guidance from the National Institute of Standards and Technology — that aim to promote fairness, transparency, and accountability in AI development and deployment.

Results and Interpretation

The analysis confirmed that algorithmic bias is not an isolated technical problem but a systemic issue rooted in data collection practices, institutional structures, and the assumptions embedded in model design. Addressing bias requires a combination of technical interventions such as bias auditing and fairness-aware training, as well as organizational commitments to inclusive development practices and ongoing monitoring after deployment.

The review of regulatory frameworks revealed a growing international consensus that AI

systems must be explainable, auditable, and subject to human oversight — particularly in high-stakes domains such as healthcare, criminal justice, and employment.

Key Takeaway

Ethical AI development is not optional. As AI systems become more embedded in consequential decisions, the responsibility to identify and mitigate bias becomes a core professional obligation for anyone working in the field.